Tanmay Patil

Data Scientist

tanmayp162004@gmail.com · +91 8080380670 · Portfolio Website · Tanmay Patil · 1Tanmay6

Data Scientist building production **LLM & agentic systems** (**RAG, fine-tuning, multi-agent orchestration**) across finance, supply chain & insurance for enterprise clients.

Experience

Data Scientist *Bangalore, KA*
Genpact *Jun 2025 – Present*

- Built a **self-improving context engine** that lets AI agents learn from feedback **without retraining** (ACE framework, ICLR 2026) – cut adaptation cost ∼**80%** vs. fine-tuning, live across 2 production environments.
- **Fine-tuned LLM** on **Azure ML** for invoice–PO matching, lifting accuracy ∼**40%** on noisy real-world vendor data.
- Built a **RAG + multi-agent demand-forecasting** system for a global CPG client, automating forecast-model selection and cutting planner workload ∼**40%**.
- Built a **synthetic data pipeline** (Meta Synthetic Data Kit + LLM-as-judge) to auto-generate and quality-filter reasoning data for downstream fine-tuning.
- Building a **multi-agent field-dispatch system** *(in progress)* for technician allocation – tool calls, DB reads/writes, and routing decisions in **under 10s** per work order at ∼**90% accuracy**.

Data Science Intern *Bangalore, KA*
Genpact *Jan 2025 – Jun 2025*

- Architected **"Data Marketplace"**, a **CTGAN + CopulaGAN + LLM** synthetic-data pipeline that cut client POC/onboarding time ∼**40%** – still in production use.
- Built a hybrid **OCR + VLM** insurance form parser, boosting field-extraction accuracy ∼**30%** over OCR-only baselines.

Data Science Intern *Remote*
Vanquisher Software Services *Jun 2024 – Nov 2024*

- Shipped a **prompt-engineering patch** for a Big-4 tax-auditing pipeline, boosting live extraction accuracy ∼**30%**.
- Built a **multi-agent education-content generator**, cutting educator prep time ∼**80%** for a national education-system client.

Education

NIIT University *Sept 2021 – June 2025*
B.Tech in Computer Science, Specialization in Data Science · GPA: **9.56/10.0**

Projects

Tabular-data-conversation (SQL Agent) *GitHub*

- Built a **graph-driven multi-pass SQL agent** on Llama 2 (pre-dating mainstream agent frameworks), boosting SQL-generation accuracy ∼**60%** over single-pass prompting on complex multi-table queries.

Automated HR Support Platform *GitHub*

- Built a **hybrid screening engine** (deterministic ATS rules + LLM evaluation) with a configurable top-K threshold to surface top talent from high-volume applicant pools.

VLM-based Security Monitoring (ESP32-CAM)

- Built a real-time smart-security system pairing an **ESP32-CAM** with a **Vision-Language Model**, generating context-aware activity updates every **10s** instead of passive motion alerts, triggered only on presence detection to cut unnecessary inference.

Technical Skills

GenAI & Agentic Systems: LangChain, LangGraph, LlamaIndex, RAG, Agentic Systems, Prompt Engineering, LLM Fine-tuning, Vision-Language Models (VLMs), Pytorch, Tensorflow, vLLM, Model Context Protocol (MCP), Hallucination Evaluation, Tool Calling
Core ML, Cloud & Tools: Python, Azure ML, AWS Bedrock, CTGAN / CopulaGAN, Meta Synthetic Data Kit